Sugandh Kumar, Ph.D.

Innovative genomic medicine researcher and computational, AI biologist
Senior Computational Biologist UCSF | Computational Genomics & AI/ML | Chemoinformatics

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Professional Summar

More than 10 years of experienced in biomedical and bioinformatic research scientist with a strong academic and industry background including five years of postdoctoral research experience at UCSF. Proven track record of designing scalable analytical pipelines, discovering clinical biomarkers, and driving precision medicine workflows. Expert in high-resolution functional genomics, translational bioinformatics, and machine learning architectures applied to complex autoimmune, inflammatory, and oncological diseases. Deep technical expertise in single-cell multi-omics integration, spatial transcriptomics, multi-ethnic GWAS, and polygenic risk scoring to map pathological cell states and therapeutic pathways. Exceptional record of peer-reviewed scholarship featuring over 30 publications, including multiple first and co-corresponding authors in high-impact manuscripts under active review or accepted. Highly collaborative and motivated to spearhead groundbreaking discoveries, with extensive experience coordinating cross-functionally across NIH-funded projects, pharmaceutical partnerships, and multidisciplinary clinical teams.

Core Competences

Single-Cell & Spatial Genomics

scRNA-seq | CITE-seq | scATAC-seq | Xenium Spatial | Visium/ST | Cell Annotation | Pseudotime | RNA velocity | Cell chat| GSEA | IPA

GWAS & Polygenic Risk Scoring

Multi-ethnic GWAS | Regression |PRS calculation & validation | MIS/TOPMed imputation | PLINK2 | vcftools | bcftools | GATK | Axiom genotyping arrays

Bulk Genomics & Epigenomics

Bulk RNA-seq (STAR, DESeq2, edgeR) | ChIP-seq | ATAC-seq | WGS/WES (GATK, ANNOVAR) | Methylation | WGCNA co-expression | Networks

Machine Learning & AI

Classical ML | Generated AI | CNN | PyTorch | TensorFlow | Feature selection | Ensemble modeling | Biomarker discovery | Treatment response prediction

Programming & Infrastructure

R | Python | Bash | Perl | HTML | CSS | JavaScript | HPCC/HPC clusters | AWS | Docker | Snakemake | Git/GitHub | Linux

Chemoinformatics & Drug Discovery

Molecular Modeling | Docking | MD simulations | QSAR/ADMET | Schrodinger Suite | Automation Tools| ROSETTA

Work Experience

Postdoctoral Research Fellow *June 2021 -- Present*

University of California, San Francisco (UCSF) -- Department of Dermatology, Liao Lab

Conducted advanced single-cell multi-omics integration and computational analysis of diverse datasets—including scRNA-seq, CITE-seq, scATAC-seq, Xenium Spatial Transcriptomics, bulk RNA-seq, ChIP-seq, WGS, and GWAS—from tissue biopsies and PBMCs to resolve immune cell infiltration pathways in chronic inflammatory skin and musculoskeletal diseases (Psoriasis, Atopic Dermatitis, Psoriatic Arthritis, and Spondyloarthritis).

Leveraged pharmacogenetics and machine learning architectures to predict patient clinical responses to biologic therapies (including anti-IL-4/IL-13 and anti-IL-23 pathways) by integrating longitudinal scRNA transcriptomic, clinical, laboratory, and genomic data from Phase III clinical trials.

Developed AI/ML-driven biomarker discovery pipelines using scRNA-seq/CITE-seq expressions and GWAS data to discover personalized therapeutic targets, aiming to improve patient outcomes and optimize clinical trial stratification between Psoriasis (PsO) and Psoriatic Arthritis (PsA) in translational dermatology.

Investigating the immunological impact of sleep architecture on chronic systemic inflammation by tracking longitudinal cell-state shifts via custom CITE-seq tracking from PBMC workflows.

Led a multi-ethnic Genome-Wide Association Study (GWAS) cohort analysis, identifying novel risk variants and elucidating complex genotype-phenotype associations to define the genetic architecture of Hidradenitis Suppurativa (HS), psoriasis, atopic dermatitis.

Architected proprietary, in-house polygenic risk score (PRS) computation engines using genetic variant arrays to quantify the cumulative statistical impact of risk alleles on patient disease susceptibility.

Directed a multimodal AI engineering project to develop a web-based clinical tool utilizing computer vision on sequential images and video feeds; automated real-time Psoriasis Area and Severity Index (PASI) scoring to objectively track longitudinal treatment efficacy.

Collaborated cross-functionally with multiple principal investigators and industry partners, ensuring timely delivery of milestones, and contributed key preliminary data to successful NIH and pharmaceutical research grants.

Provided academic leadership and mentorship to undergraduate researchers and clinical medical fellows, supervising technical bioinformatics workflows and experimental execution.

Research Associate *2015 -- 2021*

Institute of Life Sciences, Bhubaneswar, India

Cancer Biomarkers: Developed ML-based biomarkers for early-stage head and neck squamous cell carcinoma (HNSCC) using microRNA expression profiling; published in PeerJ (2020).

Multi-Omics Cancer Genomics: Conducted comprehensive genomic analyses for oral cancer including bulk RNA-seq, ChIP-seq, GATK variant calling, methylation, and metagenomic datasets.

Computational Drug Discovery: Developed an automated virtual screening pipeline for large chemical compound libraries; performed molecular docking, MD simulations, and QSAR modeling for kinase receptors and other cancer target genes for small molecules design against oral and head and neck cancer.

Comprehensive HNSCC Database: Developed a comprehensive data based for the head and neck cancer genomics expression, 4k full length model structures, millions of synthetic and natural compounds, mutational impact, etc.

Supervised and mentorship: Supervised and provided mentorship to several under Ph.D. research fellows with several Master of Sciences (M.Sc.,) fellow on computational drug designing and ML model used in biology

Collaboration initiation: Cross-functional bioinformatics collaborations with multiple principal investigators internally and across external institutions, driving computational workflows that culminated in several peer-reviewed publications.

Software Trainee (Internship) *2014 -- 2015*

Apt Software Pvt. Ltd., Kolkata, India

- Proactively delivered structural and computational expertise to influence efforts in targets and lead identification as well as lead optimization
- Developed QSAR models for predicting HIV drug activity; contributed to software testing and scripting.
- Worked on complex data analysis through R and python where analysis situation decided further molecular validation.
- Worked closely with inter-disciplinary projects teams deployed in multiple therapeutic area

Education

Ph.D., Computational Biology *2015 -- 2020*

Institute of Life Sciences, Bhubaneswar, India |

Junior & Senior Research Fellowship, University Grants Commission (UGC)

M.Sc., Bioinformatics *2011 -- 2013*

Central University of South Bihar, India | Merit Scholarship

B.Sc., Biotechnology *2008 -- 2011*

Punjab Technical University, India

Selected Publications

First-Author & Corresponding Author (Under Review / In Press)

1. Jacqueline M. Frost*, Sugandh Kumar* et al., Genetic Analysis of Pediatric Psoriasis. **[Accepted: JID, 2026 -- Co-first Author]**

2. Sugandh Kumar et al., Dupilumab Reprograms Cutaneous Immune, Neuronal, and Stromal Cells in Atopic Dermatitis. **[Under Review: JACI, 2025 -- 1st Author/Co-corresponding]**

3. Sugandh Kumar et al., Cellular Indexing of Transcriptomes and Epitopes (CITE-Seq) in Hidradenitis Suppurativa identifies dysregulated cell types and facilitates ML-based diagnosis. **[Under Review: Nature Communications, 2025 -- 1st Author/Co-corresponding]**

Published Peer-Reviewed Articles (Selected)

4. Faye Orcales, **Sugandh Kumar** et al., A partitioned polygenic risk score reveals distinct contributions to psoriasis clinical phenotypes across a multi-racial cohort. Journal of Translational Medicine, 2024.
5. Bahram Razani, **Sugandh Kumar** et al., A20 (Tnfaip3) in keratinocytes restricts an early antiviral gene program associated with psoriatic skin and joint disease. JID, 2024.
6. Gian Carlo, **Sugandh Kumar** et al., Immunological differences in Atopic Dermatitis across age groups: Insights from single-cell multi-omics. Journal of Translational Medicine, 2026 (Accepted).
7. Antonia Wiala, **Sugandh Kumar** et al., Patients with Psoriasis and Suppurative Hidradenitis (PSOSH) share genetic risk factors. JAAD, 2025.
8. Jared Liu, **Sugandh Kumar** et al., combining single-cell transcriptome and surface epitope profiling with ML to diagnose psoriatic arthritis. Frontiers in Immunology, 2022.
9. Audrey Bui, **Sugandh Kumar** et al., A weighted genetic risk score using 88 susceptibility variants in Newfoundland psoriasis patients. Frontiers in Genetics, 2023.
10. **Sugandh Kumar**, Srinivas Patnaik, Anshuman Dixit, Predictive models for stage and risk classification in HNSCC. PeerJ, 2020.

Full list of publication : Google Scholar: *More than 30 peer-reviewed articles, 5 book chapter author, 1 Australian innovation patent (No. 2021103325). Active reviewer 200+ manuscript of 30+ journals, journals editors across 6 journals.*

Awards, Recognition & Service

- Selected for SID world leadership in Dermatology and Immunology Conference - 2026
- NIH Grant Update Lecture, UCSF Immunology Conference (invited speaker) - 2022
- Best Poster Award -- Genomics India 2019 International Conference in Human Genomics
- Junior & Senior Research Fellowship, University Grants Commission (UGC), Government of India (2015-2020)
- Journal Editor: Frontiers in Bioinformatics (2023); Guest Editor: MDPI Biomolecules (2023)
- Active reviewer: Scientific Reports, Cell Reports, Frontiers in Pharmacology, MDPI Journals (100+ reviews)
- Australian Innovation Patent No. 2021103325 -- Method to Detect Paternal Contributors in Seminal Extracellular Vesicles for Recurrent Pregnancy Loss

~ References and other documents available upon request